Counting a Nested Loop

Topic: TimeComplexity File: p2.pdf

PROBLEM

Determine the time complexity of two nested loops, each executing n times.

PSEUDOCODE

```
for i = 0 to n - 1:
    for j = 0 to n - 1:
        do_constant_work()
```

ANALYSIS

For each of the n values of i, the inner loop performs n iterations. Total work is n times n, or n^2.

EXAMPLE

When n = 3, the inner statement runs 3 * 3 = 9 times. When n doubles to 6, it runs 36 times.

COMPLEXITY

Time: O(n^2). Extra space: O(1), assuming the loop body uses constant additional memory.

TAKEAWAY

Nested loops are not automatically quadratic; count how often the inner work actually runs.